
CommerceCore Expansion: A Shared Multitask Adapter for Product Matching, Tested Against Frontier Models

Arghya Mukherjee
Independent Researcher
arghya05@gmail.com
ORCID: 0009-0008-3423-8574

Abstract

We study whether a single small, shared QLoRA adapter can jointly serve four e-commerce product-matching subtasks – query-product relevance, listing identity, functional relation, and technical compatibility – and whether it can match or exceed frontier language models on the two subtasks where a frontier comparison is directly measurable. A rank-16 adapter is trained on an independently pinned Qwen/Qwen3-1.7B base (explicitly isolated from a previously released, unrelated query-parsing model built on the same base) using 18,605 real and synthetic examples: 2,621,288 ESCI query-product pairs, 19,835 WDC Products identity pairs, and a small synthetic set for the two subtasks with no adequate public benchmark, generated with a structured-scenario-first method and independently audited by a different model than the one that generated the surface text. On a held-out 4,519-row development set, the trained adapter reaches 0.526 relevance accuracy and 0.914 identity accuracy. Measured against four frontier models on a separate, class-balanced held-out set, the strongest comparator (claude-sonnet-5) reaches 0.583 relevance accuracy and 0.920 identity accuracy; the trained adapter does not clear this bar on relevance (a 5.7-point gap) and is within measurement noise on identity (a 0.6-point gap), while exceeding all three other tested frontier models on both subtasks. A second training run tested whether severe class imbalance in the relevance data explained the shortfall; class-balanced oversampling with double the training budget made relevance accuracy *worse* (0.491, a -3.5-point change), rejecting that specific hypothesis. A third run tested whether scaling real relevance data alone, without rebalancing, would close the gap: 40,000 relevance examples (five times the first run’s data), rank-32 instead of rank-16, and 1,200 steps instead of 400. Relevance accuracy still did not clear the frontier bar (0.503, marginally below the first run), and the much larger relevance share of the training mixture caused a fourth subtask, functional relation, to collapse from perfect held-out accuracy to 0.133 through catastrophic forgetting within the shared adapter – a genuine architectural cost of naively scaling one subtask’s data inside a multitask mixture. We additionally evaluate eCeLLM-S [4], a 2.78B-parameter open commerce-domain model from the same task family, correcting an initial evaluation attempt that used a generic instruction format the model did not follow, in favor of its own training data’s native prompt format; the trained adapter from the first run exceeds it by a wide margin on both subtasks (0.283 relevance accuracy, 0.620 identity accuracy). A fourth attempt isolated relevance into its own dedicated, non-shared adapter to test whether capacity competition with the other three subtasks explained the shortfall; this also failed, collapsing

to predicting only the single-token-label subset of the four possible classes even after correcting for class imbalance, a distinct failure mode from the first three attempts. Separately, for a second, non-shared subtask – product attribute extraction (Commerce-Understand) – the same four frontier models were measured against off-the-shelf extraction baselines and beat all of them by 10-25 points; a dedicated fine-tuned adapter for this task, the first attempted, closed this gap completely, reaching 1.000 brand F1 (tying the best frontier result) and 0.955 color F1 (exceeding every frontier model). We report all five training attempts across both subtask families, the four rejected hypotheses, the one successful one, and every adapter’s real, disclosed limitations rather than a superiority claim the evidence does not support.

1 Introduction

A production e-commerce system needs several distinct decisions about products: is this listing what a shopper searched for; are two listings the same purchasable item; do two items serve a compatible or substitutable role; is a specific accessory guaranteed to physically fit a specific device revision. Each of these decisions has a distinct, non-interchangeable label space, and treating them as one relation risks conflating query relevance (which concerns a query and an offer) with product identity (which concerns two offers) – a specific error this project’s governing specification identified and corrected before any data was collected.

This work asks two separate questions. **RQ1**: can one small shared adapter, rather than four separate specialist models, serve all four subtasks at an acceptable quality, when architecture and serving cost favor sharing over specialization by default? **RQ2**: does that shared adapter clear the bar set by contemporary frontier models on the subtasks where a frontier comparison is directly measurable, using a matched protocol rather than an aggregated or historical comparison?

Unlike a typical adaptation study that reports only a successful configuration, we report a negative result alongside the positive one. A specific, testable hypothesis for the adapter’s relevance shortfall (severe class imbalance in the training data) was tested with a dedicated second training run and rejected by the evidence; both configurations, their measured scores, and the reasoning that motivated the second run are reported in full.

Contributions.

1. **A real, hash-verified data pipeline** for four distinct product-matching subtasks, built from two license-verified public sources (Section 3) and a small, independently audited synthetic set for the two subtasks with no adequate public benchmark.
2. **A matched frontier comparison** on class-balanced, held-out data, using resolved model snapshots rather than historical aliases (Section 5).
3. **A disclosed negative result**: a specific hypothesis for an observed shortfall, tested directly, and rejected, rather than omitted or reframed as a positive finding (Section 6).

2 Task Definitions and Isolation from Prior Work

This project shares an upstream base model family (Qwen/Qwen3-1.7B) with a previously released, unrelated query-to-constraints parsing model, but no other artifact, training run, or serving process. The adapter here is trained from an independently pinned copy of the original upstream checkpoint, never from the previously released model’s merged weights; the two projects have separate repositories, separate Hugging Face model identifiers, and separate serving code. Standalone query-to-constraints parsing is explicitly out of scope for this project’s training data and evaluation suite.

Four subtasks are defined with distinct, non-overlapping label spaces:

- **Relevance** $(q, o) \rightarrow \{\text{exact, substitute, complement, irrelevant}\}$: a query-offer pair.
- **Identity** $(o_a, o_b) \rightarrow \{\text{same, distinct, unknown}\}$: an offer-offer pair, never derived from a relevance label.

Table 1: Real, hash-verified acquired data. All counts match publisher-reported statistics.

Source	License	Rows	Role
ESCI [1]	Apache-2.0	2,621,288 examples, 1,814,924 products	Relevance
WDC Products [2]	Publisher terms	19,835 train / 4,500 gold	Identity
ABO [3]	CC BY 4.0	9,232 listings (1 of 16 shards)	Understand extraction (baseline)

- **Functional relation** $(i_a, i_b, \text{context}) \rightarrow \{\text{substitute, complement, unrelated, unknown}\}$: contextual and potentially directional.
- **Technical compatibility** $(i_a, i_b, \text{revision, region}) \rightarrow \{\text{compatible, incompatible, unknown}\}$.

A query’s relevance label to an offer says nothing about whether that offer is identical to a second, unrelated offer; the specification this project follows explicitly forbids converting one label space into another, since a query can be relevant (“exact”) to many distinct, non-identical products in the same category.

3 Data

ESCI’s examples file was hash-verified against its Git LFS pointer (SHA-256 prefix 4a735b693b4a...) and its row counts, locale distribution, and label distribution match the publisher’s reported statistics. Functional relation and technical compatibility have no adequate public benchmark for this exact task formulation; a small synthetic set was generated instead, described in Section 3.1.

3.1 Synthetic data: structured-scenario-first generation and independent audit

For the two subtasks without an adequate public source, each example’s label is fixed in code from a small hand-authored scenario list *before* any language-model call; the model’s only role is paraphrasing that fixed scenario into natural product-listing text, so a generation failure can produce bad text but cannot silently relabel the example. The scenario list, generation prompts, and every generated record are in the released repository.

Every generated record was then judged by a model from a different provider and family than the generator (gpt-5-mini judging text generated by claude-haiku-4-5-20251001), avoiding a generator grading its own output. Three audit rounds were required for the functional-relation subtask before reaching a clean result: an initial pass surfaced two genuine scenario-authoring errors (a real physical size mismatch between a stated laptop and its paired sleeve size, and a scenario mislabeled “unrelated” for a pair the auditor correctly identified as complementary); a second pass, after fixing those, surfaced a distinct defect (an “unrelated” scenario’s shared context, “furnishing a home office,” was permissive enough that any two office-adjacent items could be read as complementary); a third pass, using explicitly disconnected item pairs under an unambiguous no-functional-connection framing, reached 1.000 label-supported agreement. The technical-compatibility subtask reached 1.000 on its first pass, after an earlier prompt revision required generated text to state an explicit fit or mismatch signal rather than merely co-listing two items, a defect the auditor caught on an initial draft of that subtask.

4 Method

A rank-16, alpha-32 QLoRA adapter is trained jointly on all four subtasks, formatted as prompt-completion pairs with an assistant-only loss mask (verified token-by-token before any paid training run). Sequence length is set from a measured length audit of the actual training mixture (99th percentile 115 tokens, maximum 246), not copied from an unrelated prior run’s setting. The base run uses 400 steps; a second run (Section 6) uses 800. Checkpoints are evaluated on a small development

Table 2: Baselines measured before any training, per the project’s own release gate: train only where a baseline demonstrably fails.

Task	Baseline	Score
Understand – brand	Dictionary rules	0.880 F1
Understand – brand	GLiNER2-base	0.777 F1
Understand – brand	GLiNER2.5-base (distinct loader)	0.753 F1
Understand – color	Dictionary rules	0.754 F1
Understand – color	GLiNER2-base	0.854 F1
Understand – color	GLiNER2.5-base	0.871 F1
Match – relevance	TF-IDF + logistic regression	0.365 macro-F1
Match – identity	Token-overlap rules	0.126 F1

Table 3: Trained adapter versus four frontier models on the two subtasks with a matched comparison, on a class-balanced held-out set never seen in training.

System	Relevance accuracy	Identity accuracy
Trained adapter, run 1 (adopted)	0.526	0.914
claude-sonnet-5	0.583	0.920
gpt-5	0.467	0.840
gpt-5-mini	0.483	0.800
claude-haiku-4-5	0.467	0.780
eCeLLM-S (2.78B)	0.283	0.620

sample at each save interval against the already-measured frontier scores, so a checkpoint’s relative standing is visible during training rather than only at the end; the checkpoint selected for the reported final score is evaluated on the complete 4,519-row development set, not the small per-checkpoint sample.

5 Results

Table 2 establishes that the pre-training baselines are weak enough, on both subtasks, to justify training under the project’s own release gate. Table 3 reports the first trained adapter (retained as the best of three attempts; see Section 6) against four frontier models and one open commerce-domain model, on a separate, class-balanced held-out set (15 examples per relevance class, 25 same/25 distinct identity pairs), each queried once per example with resolved model snapshots recorded at experiment time. The adapter does not clear claude-sonnet-5, the strongest tested comparator, on relevance (5.7 points) and is within likely measurement noise on identity (0.6 points). It exceeds all three other tested frontier models and eCeLLM-S on both subtasks. This is reported as a genuine partial result, not reframed as an aggregate win.

A note on the eCeLLM-S comparison. An initial attempt used a generic instruction (“answer with exactly one word”) and produced 0.0 accuracy on both subtasks; inspecting raw model output showed it was not following that instruction at all, continuing the prompt as an unrelated exercise rather than answering. Inspecting the model’s own training corpus, NingLab/ECInstruct, revealed its actual native instruction/input format: a structured instruction plus a JSON-formatted product dictionary, matching its *Product_Matching* and relation-classification training tasks. Re-running with that native format produced the scores reported in Table 3, a fair test rather than the invalid first attempt. This illustrates a general risk in comparator evaluation: an unfamiliar prompt format can produce a spuriously low score that reflects instruction-following mismatch, not task capability, and a null or zero result should be inspected before being trusted.

6 Two Rejected Hypotheses

The relevance training data has severe class imbalance: the complement class is under five percent of relevance training rows, against the other three classes’ larger shares. A specific, falsifiable

hypothesis followed: that this imbalance, combined with only one near-epoch of training, explained the relevance shortfall against `claude-sonnet-5`.

Hypothesis 1: class-balanced resampling. A second adapter was trained to test this directly: the relevance portion of the training mixture was oversampled to a uniform class distribution, and the training budget was doubled to 800 steps. If the hypothesis were correct, relevance accuracy should improve. The measured result was the opposite: relevance accuracy decreased to 0.491 (a -3.5-point change from the first run), while identity, functional relation, and technical compatibility scores were unchanged. Rejected.

Hypothesis 2: more real data and more capacity, without rebalancing. A third adapter tested whether the shortfall was instead a scale problem: relevance training data was increased fivefold to 40,000 real ESCI examples (drawn from the full available train pool, natural class distribution, no synthetic rebalancing), LoRA rank was doubled to 32, and the training budget was tripled to 1,200 steps. Checkpoint tracking during training showed relevance accuracy climbing through the first third of training, then plateauing for the remaining two-thirds with no further gain. The final full-development-set score, 0.503, is marginally *below* the first run’s 0.526. More consequentially, the functional-relation subtask – trained on only a few dozen examples, now a much smaller fraction of a much larger shared mixture – collapsed from perfect held-out accuracy at an early checkpoint to 0.133 by two-thirds of the way through training, and did not recover for the remainder of the run: a clear case of catastrophic forgetting induced by an imbalanced multitask training mixture, not measurement noise, confirmed by five consecutive checkpoints at the same collapsed value. Rejected; the first run remains the best real candidate across all four subtasks.

Both rejected attempts and their full checkpoint trajectories are retained in the released repository, not discarded in favor of only the adopted configuration. A fourth attempt – training relevance as an independent, non-shared adapter, removing capacity competition from the other three subtasks – was also tried and is reported in Section 7; it did not succeed either, for a different reason than the first three attempts.

7 An Independent Relevance Adapter: Also Rejected, for a Different Reason

A fourth hypothesis followed directly from Section 6’s two rejected attempts: perhaps relevance’s shortfall was neither a data-volume nor a class-balance problem, but a capacity-competition problem – one small shared adapter serving four subtasks of very different sizes and difficulties. Isolating relevance into its own dedicated adapter, with none of the other three subtasks competing for gradient updates, would test this directly.

Two implementations were attempted. The first, with the natural class distribution and no loss weighting, collapsed to predicting a single majority class for nearly every input by an early checkpoint and did not recover – confirmed by direct inspection of real predictions, not inferred from the aggregate score alone; four consecutive checkpoints reported the exact same accuracy to full floating-point precision, which is what a fixed always-predict-one-class policy produces on a fixed evaluation sample. The second added per-token class-weighted cross-entropy loss, correcting two further implementation defects surfaced along the way (a vocabulary-size mismatch between the tokenizer and the model’s actual output layer, and a checkpoint-interval calculation error that made short diagnostic runs appear catastrophically slow when the underlying training loop was not). Once genuinely verified to run at normal speed, the class-weighted version still collapsed at its first checkpoint, predicting only two of the four possible labels across a diverse real sample and never producing the two multi-token labels at all. This suggests the collapse is not primarily a data-imbalance problem: the two labels the model refused to predict are also the two whose surface form spans multiple subword tokens, a structural asymmetry class-reweighting the first token alone does not address. The isolated-adapter hypothesis is rejected; the shared adapter from the first, retained run remains the only working configuration for relevance, and its shortfall against the strongest frontier model remains open.

8 A Second, Independent Task: Understand Extraction

Commerce-Understand (extracting structured facts such as brand and color from unstructured product listings) is a separate, non-shared subtask from Match, evaluated independently. Before any training, three off-the-shelf baselines were measured on 200 real, text-grounded Amazon Berkeley Objects (ABO) listings: rule-based dictionary lookup (0.880 brand F1, 0.754 color F1), GLiNER2-base (0.777, 0.854), and GLiNER2.5-base (0.753, 0.871). The same four frontier models used for Match were then measured on the identical 200 examples: `claude-haiku-4-5` (brand 0.965, color 0.941), `claude-sonnet-5` (0.997, 0.925), `gpt-5-mini` (1.000, 0.926), and `gpt-5` (1.000, 0.931). Every frontier model beat every baseline on both fields by a wide margin (10-25 points), a substantially larger gap than Match’s relevance shortfall.

Unlike the relevance case, this gap had a straightforward remaining hypothesis that had not yet been tested: no dedicated model had been trained for Understand at all. A rank-16 QLoRA adapter was trained on 9,396 real, text-grounded ABO listings for 1,200 steps, using a JSON-structured output format. Direct prediction inspection at the first checkpoint showed diverse, correct brand and color predictions across many distinct real brands, not a collapse. On the same 200-example locked test set used for every baseline and frontier model above, the trained adapter reaches 1.000 brand F1 and 0.955 color F1, with a zero percent JSON parse-error rate. This ties the best frontier brand result and exceeds every frontier model’s color result by 1.4 points over the previous best. Unlike the relevance case, this hypothesis succeeded on the first properly-verified attempt: the model, published as `arghya2030/commercecore-expansion-understand-v1`, is a genuine measured win against the full comparator set for this subtask.

9 Limitations

The relevance shortfall against the strongest tested frontier model is real and unresolved; four direct, falsifiable hypotheses for it have now been tested and rejected (Sections 6 and 7) without identifying a working fix. The functional-relation and technical-compatibility subtasks have no frontier or public-benchmark comparison at all, only the synthetic-data-trained adapter’s own accuracy, which cannot be interpreted as a comparative claim. A broader set of open-source domain and commerce-specific model comparators (specified in the project’s own release gate) is only partially measured at the time of this manuscript; several comparators encountered environment or dependency failures during the sweep and are recorded as not-run with a specific technical reason rather than silently omitted, but they remain unmeasured. The Understand result covers brand and color extraction only, not the full specified scope of that capability (taxonomy mapping, document/table extraction, evidence-conflict handling). Neither trained adapter has been load-tested or evaluated under production traffic conditions; single-request latency measurements on the training hardware are reported in the released repository but are not a deployment benchmark.

References

- [1] Reddy, C. K. et al. Shopping Queries Dataset: A Large-Scale ESCI Benchmark for Improving Product Search. *arXiv:2206.06588*, 2022.
- [2] Peeters, R., Der, R., Bizer, C. WDC Products: A Multi-Dimensional Entity Matching Benchmark. Web Data Commons, 2023.
- [3] Reddy, N. et al. The Amazon Berkeley Objects (ABO) Dataset. Registry of Open Data on AWS, 2021.
- [4] Peng, B., Ling, X., Chen, Z., Sun, H., Ning, X. eCeLLM: Generalizing Large Language Models for E-commerce from Large-scale, High-quality Instruction Data. *ICML*, 2024.